

[시뮬레이션 순서] \* 첨부한 zip 파일 참고

## 0. zip 파일 풀기

### 1. fftool, packmol 을 이용하여 다음 시스템을 만든다

- (1) L1P1 (LiFSI 1/ Pyr13FSI 1) = Li 50 개 + Pyr13 50 개 + FSI <sup>100</sup> 50 개  
(2) L1P2 (LiFSI 1/ Pyr13FSI 2) = Li 50 개 + Pyr13 100 개 + FSI 150 개  
(3) L2P1 (LiFSI 2/ Pyr13FSI 1) = Li 100 개 + Pyr13 50 개 + FSI 150 개  
(4) L3P1 (LiFSI 3/ Pyr13FSI 1) = Li 150 개 + Pyr13 50 개 + FSI 200 개

예, L1P1

#### (1) typing

fftool 50 Li.zmat 50 <sup>100</sup> c3c1pyrr.zmat 50 <sup>100.0</sup> fsi.zmat -b 10.0 <sup>10 Å</sup> (box size = 10 Å each)

Output:

density 249.087 mol/L volume 1000.0 Å<sup>3</sup>

molecule descriptions

Li.zmat

c3c1pyrr.zmat

fsi.zmat

| species   | nmol | bonds | charge  |
|-----------|------|-------|---------|
| Li+       | 50   | none  | +1.0000 |
| c3c1pyrr+ | 50   | none  | +1.0000 |
| fsi-      | 50   | none  | -1.0000 |

packmol file

pack.inp

(2) typing

packmol < pack.inp

pack mol  
(2)

Output:

(중간 생략)

The solution with the best function value was  
written to the output file: simbox.xyz

---

Forcing the solution to fit the constraints...

---

The forced point was written to the  
output file: simbox.xyz\_FORCED

If you want that the packing procedure continues  
for a longer time, add the following keyword  
to the input file:

nloop [integer]      (ex: nloop 200)

The default nloop value is 50 for each molecule.

#####  
#####

ENDED WITHOUT PERFECT PACKING:

The output file:

simbox.xyz

contains the best solution found.

Very likely, if the input data was correct,

it is a reasonable starting configuration.

Check commentaries above for more details.

#####  
#####

Running time: 881.745422 seconds.

(3) typing

fftool 50 Li.zmat 50 c3c1pyrr.zmat 50 fsi.zmat -b 10.0 -g (g: gromacs input 파일 생성)

Output:

density 249.087 mol/L volume 1000.0 A^3

molecule descriptions

Li.zmat

c3c1pyrr.zmat

fsi.zmat

| species   | nmol | bonds | charge  |
|-----------|------|-------|---------|
| Li+       | 50   | file  | +1.0000 |
| c3c1pyrr+ | 50   | file  | +1.0000 |
| fsi-      | 50   | file  | -1.0000 |

gromacs files

run.mdp

field.top

config.pdb

## 2. gromacs (linux 버전 또는 windows 버전) 설치하기

## 3. gromacs 실행 (minimization) : 50000 steps

```
gmx grompp -f run0.mdp -c config.pdb -p field.top -o run0.tpr -maxwarn 5
```

```
gmx mdrun -v -ntmpi 1 -deffnm run0 -pin on
```

## 4. gromacs 실행 (equilibration 1) : 1 ns

```
gmx grompp -f run1.mdp -c run0.gro -p field.top -o run1.tpr -maxwarn 5
```

```
gmx mdrun -v -ntmpi 1 -deffnm run1 -pin on
```

## 5. gromacs 실행 (equilibration 2) : 10 ns

```
gmx grompp -f run2.mdp -c run1.gro -r run1.gro -t run1.cpt -p field.top -o run2.tpr -
```

maxwarn 5

gmx mdrun -v -ntmpi 1 -deffnm run2 -pin on

## 6. gromacs 실행 (dynamics run 또는 production run) : 10 ns

gmx grompp -f run3.mdp -c run2.gro -r run2.gro -t run2.cpt -p field.top -o run3.tpr -  
maxwarn 5

gmx mdrun -v -ntmpi 1 -deffnm run3 -pin on

## 7. 시뮬레이션 데이터 해석하기

(0) 이온의 특정 원자를 택하기 위해서 index file 을 만들어야 함

gmx make\_ndx -f run3.gro -o index.ndx

Output:

:-) GROMACS - gmx make\_ndx, 2021.7 (-:

GROMACS is written by:

|                     |                       |                  |
|---------------------|-----------------------|------------------|
| Andrey Alekseenko   | Emile Apol            | Rossen Apostolov |
| Paul Bauer          | Herman J.C. Berendsen | Par Bjelkmar     |
| Christian Blau      | Viacheslav Bolnykh    | Kevin Boyd       |
| Aldert van Buuren   | Rudi van Drunen       | Anton Feenstra   |
| Gilles Gouaillardet | Alan Gray             | Gerrit Groenhof  |
| Anca Hamuraru       | Vincent Hindriksen    | M. Eric Irrgang  |
| Aleksei Iupinov     | Christoph Junghans    | Joe Jordan       |
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| Pascal Merz        | Pieter Meulenhoff | Teemu Murtola    |
| Szilard Pall       | Sander Pronk      | Roland Schulz    |
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| Peter Tieleman     | Jon Vincent       | Teemu Virolainen |
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and the project leaders:

Mark Abraham, Berk Hess, Erik Lindahl, and David van der Spoel

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GROMACS: `gmx make_ndx`, version 2021.7

Executable: `/usr/local/gromacs/bin/gmx`

Data prefix: `/usr/local/gromacs`

Working dir: `/home/users/schoe/projects/IL_shanjida/02_test_RandE`

Command line:

```
gmx make_ndx -f run3.gro -o index.ndx
```

Reading structure file

Going to read 0 old index file(s)

Analysing residue names:

There are: 300 Other residues

Analysing residues not classified as Protein/DNA/RNA/Water and splitting into groups...

0 System : 4100 atoms

1 Other : 4100 atoms

2 Li+ : 50 atoms

3 c3c : 2700 atoms

4 fsi : 1350 atoms

nr : group '!' : not 'name' nr name 'splitch' nr Enter: list groups

'a': atom '&': and 'del' nr 'splitres' nr 'l': list residues

't': atom type '|' : or 'keep' nr 'splitat' nr 'h': help

'r': residue 'res' nr 'chain' char

"name": group 'case': case sensitive 'q': save and quit

'ri': residue index

> a OBT

Found 600 atoms with name OBT

5 OBT : 600 atoms

> q

GROMACS reminds you: "Statistics: The only science that enables different experts using the same figures to draw different conclusions." (Evan Esar)

(위와 같이 FSI 을 이루고 있는 산소 원자 OBT 의 index 를 추가할 수 있음)

(1) radial distribution function (rdf) 구하기 (예, Li vs. O(FSI))

```
gmx rdf -f run3.xtc -s run3.tpr -n index.ndx -o rdf_Li_OFSI.xvg
```

Output:

:-) GROMACS - gmx rdf, 2021.7 (-:

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GROMACS: gmx rdf, version 2021.7

Executable: /usr/local/gromacs/bin/gmx

Data prefix: /usr/local/gromacs

Working dir: /home/users/schoe/projects/IL\_shanjida/02\_test\_RandE

Command line:

```
gmx rdf -f run3.xtc -s run3.tpr -n index.ndx -o rdf_Li_OFSI.xvg
```

Available static index groups:

Group 0 "System" (4100 atoms)

Group 1 "Other" (4100 atoms)

Group 2 "Li+" (50 atoms)

Group 3 "c3c" (2700 atoms)

Group 4 "fsi" (1350 atoms)

Group 5 "OBT" (600 atoms)

Specify a selection for option 'ref'

(Reference selection for RDF computation):

(one per line, <enter> for status/groups, 'help' for help)

> 2

Selection '2' parsed

Available static index groups:

Group 0 "System" (4100 atoms)

Group 1 "Other" (4100 atoms)

Group 2 "Li+" (50 atoms)

Group 3 "c3c" (2700 atoms)

Group 4 "fsi" (1350 atoms)

Group 5 "OBT" (600 atoms)

Specify any number of selections for option 'sel'

(Selections to compute RDFs for from the reference):

(one per line, <enter> for status/groups, 'help' for help, Ctrl-D to end)

> 5

Selection '5' parsed

> Reading file run3.tpr, VERSION 2021.1 (single precision)

Reading file run3.tpr, VERSION 2021.1 (single precision)

Last frame 5000 time 50000.000

Analyzed 5001 frames, last time 50000.000

GROMACS reminds you: "Let Me Do This" (Urban Dance Squad)

(첫번째 2 를 입력하여 Li 를 선택하고 5를 입력하여 O(FSI) 를 선택한 다음, ctrl-D 를 누르면 gromacs 가 rdf 를 계산하여 rdf\_Li\_OFSl.xvg 파일에 저장. 엑셀로 그래프를 만들 수 있는데 첫번째 컬럼이 거리  $r$ 을 나타내고 두번째 컬럼이 radial distribution function  $g(r)$  을 나타냄. 위의 방법으로 Li 이온과 특정 원자사이의 rdf 를 그릴 수 있음)

(2) coordination number 구하기 (예, Li vs. O(FSI))

```
gmx rdf -f run3.xtc -s run3.tpr -n index.ndx -o rdf_Li_OFSl.xvg -cn cn_Li_OFSl.xvg
```

(rdf 구하면서 동시에 -cn 옵션으로 coordination number 도 같이 구하고 그래프를 그릴 수 있음)

(3) msd & diffusion 구하기

```
gmx msd -f run3.xtc -s run3.tpr -n index.ndx -o msd_Li.xvg
```

Output:

:-) GROMACS - gmx msd, 2021.7 (-:

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Working dir: /home/users/schoe/projects/IL\_shanjida/02\_test\_RandE

Command line:

```
gmx msd -f run3.xtc -s run3.tpr -n index.ndx -o msd_Li.xvg
```

Reading file run3.tpr, VERSION 2021.1 (single precision)

Reading file run3.tpr, VERSION 2021.1 (single precision)

Select a group to calculate mean squared displacement for:

|       |     |         |     |      |          |
|-------|-----|---------|-----|------|----------|
| Group | 0 ( | System) | has | 4100 | elements |
| Group | 1 ( | Other)  | has | 4100 | elements |
| Group | 2 ( | Li+)    | has | 50   | elements |
| Group | 3 ( | c3c)    | has | 2700 | elements |